

Definition and first terms

$13A(x) = 12 + 27x + A(x)^4$, with the origin-normalized solution.
 Normalization/grammar: $A(x)=1+(3)*T(x)$; $T=x+2*T^2+4*T^3+3*T^4$.
 $a(0), \dots, a(7) = 1, 3, 6, 36, 249, 1932, 16044, 139500$.
 Kernel used throughout: $\rho(u) = u(1 - 2u^1 - 4u^2 - 3u^3)$, so $x = \rho(T(x))$.

Typogeometry

colored arity geometry; node color distinguishes constructors. Filled endpoints are true leaves; open endpoints are false/empty leaves. For colored grammars, color is genuine constructor data and is not silently converted into a positional zero pattern.

Typogeometric constructor codes and multiplicities: Δ_2 : 2, Δ_3 : 4, Δ_4 : 3.

Coefficient and contour extraction

$$\mathcal{K}_n = \{(k_1, \dots, k_3) \in \mathbb{Z}_{\geq 0}^3 : \sum_{j=1}^3 j k_j = n - 1\}, \quad K = \sum_{j=1}^3 k_j,$$

$$a(n) = \frac{3}{n} \sum_{\mathbf{k} \in \mathcal{K}_n} \binom{n + K - 1}{K} \binom{K}{k_1, \dots, k_3} 2^{k_1} 4^{k_2} 3^{k_3}.$$

$$a(n) = \frac{3}{2\pi i n} \oint_{\gamma} \frac{du}{\rho(u)^n}, \quad n \geq 1.$$

Recurrence

$$\sum_{r=0}^3 P_r(n) a(n+r) = 0, \quad n \geq 1, \text{ with}$$

$$\begin{aligned}
 P_0(n) &= -6912\,n^3 - 10368\,n^2 - 1296\,n + 1080 \\
 P_1(n) &= -9216\,n^3 - 27648\,n^2 - 26688\,n - 8256 \\
 P_2(n) &= -4096\,n^3 - 18432\,n^2 - 26624\,n - 12288 \\
 P_3(n) &= 451\,n^3 + 2706\,n^2 + 4961\,n + 2706
 \end{aligned}$$

Differential equation

$$\begin{aligned}
 &(1080\,x^3)\,A(x) + (-18576\,x^4 - 8256\,x^3)\,A'(x) \\
 &+ (-31104\,x^5 - 27648\,x^4 - 6144\,x^3)\,A^{(2)}(x) + (-6912\,x^6 - 9216\,x^5 - 4096\,x^4 + 451\,x^3)\,A^{(3)}(x) = 0.
 \end{aligned}$$

Matrices, telescoper certificate, and checks

$$U = \begin{pmatrix} \frac{3376}{451} & \frac{676}{451} & \frac{52}{451} & \frac{4}{451} \\ \frac{8736}{451} & \frac{672}{451} & \frac{468}{451} & \frac{36}{451} \\ \frac{8112}{451} & \frac{624}{451} & \frac{48}{451} & \frac{420}{451} \\ 0 & 0 & 0 & 0 \end{pmatrix}, \quad V = \begin{pmatrix} -1 & 0 & 0 & 0 \\ \frac{1572}{451} & \frac{225}{451} & \frac{52}{451} & \frac{4}{451} \\ \frac{260}{41} & \frac{20}{41} & \frac{11}{41} & \frac{4}{41} \\ \frac{2028}{451} & \frac{156}{451} & \frac{12}{451} & \frac{105}{451} \end{pmatrix}, \quad J = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 3 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

Certificate.

$$R(n, u) = N(n, u)/\rho(u)^2, \quad \deg_u N = 9.$$

$$\text{Telescoping identity: } \sum_r P_r(n) H_{n+r}(u) = \partial_u (R(n, u) H_n(u)).$$

Matrix dimensions: 8×8 ; remainder 3×4 , rank 3, nullity 1. The exact telescoping residual is zero. The recurrence reconstructs all 24 stored terms; 6/6 published OEIS prefix terms match.